

Gabor Granular Synthesis & Time-Stretching Math

Mathematical Models of the [sampler~] Granular Engine

Time Dilation DAW (v0.0.1) Engineering Group

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Abstract

This paper formulates the Gabor granular time-stretching algorithms, grain window envelopes, stochastic Poisson grain distributions, and overlap-add (OLA) sum equations powering the [sampler~] audio node.

1 Gabor Grain Representation

A single audio grain $g_k(t)$ of length L_k centered at buffer offset τ_k with pitch playback speed S_k is defined by modulating an audio sample buffer $x(t)$ with a smooth window envelope $w(t)$:

$$g_k(t) = w\left(\frac{t - t_{k,\text{start}}}{L_k}\right) \cdot x(\tau_k + S_k \cdot (t - t_{k,\text{start}})) \quad (1)$$

where $t_{k,\text{start}}$ is the physical time instant when grain k is spawned.

2 Grain Windowing Envelope Functions

To eliminate boundary click artifacts at grain edges ($t = 0$ and $t = L_k$), window function $w(\eta)$ for $\eta \in [0, 1]$ is selected:

2.1 Hanning Window Envelope

$$w_{\text{Hanning}}(\eta) = \frac{1}{2} (1 - \cos(2\pi\eta)) = \sin^2(\pi\eta) \quad (2)$$

2.2 Gaussian Window Envelope

$$w_{\text{Gaussian}}(\eta) = \exp\left(-\frac{(\eta - 0.5)^2}{2\sigma^2}\right), \quad \text{where } \sigma \approx 0.16 \quad (3)$$

3 Stochastic Poisson Grain Spawning Distribution

The inter-grain arrival interval Δt_{spawn} is governed by a Poisson point process with mean grain density rate λ (grains per second):

$$P(\Delta t_{\text{spawn}} = \delta) = \lambda e^{-\lambda\delta} \quad (4)$$

In discrete time, next grain spawn time n_{next} is randomized using uniform random variate $U \in (0, 1)$:

$$n_{\text{next}} = n_{\text{current}} + \left\lceil -\frac{f_s}{\lambda} \ln(U) \right\rceil \quad (5)$$

4 Overlap-Add (OLA) Grain Summation

The continuous output signal $y(t)$ is generated by summing all active overlapping grains $g_k(t) \in \mathcal{G}_{\text{active}}$:

$$y(t) = \sum_{k \in \mathcal{G}_{\text{active}}} g_k(t) \quad (6)$$

Under uniform grain overlap with Hanning windowing, constant power energy normalization $\sum w_k(t)^2 = 1.0$ is maintained, preserving original dynamic loudness across variable grain densities.